BSc (Hons) in Information Technology Specializing Data Science

Research Project - IT4010 **MLOPs Report**

Group ID: **25-26J-123**

Project Title: **AI-POWERED DATA ANONYMIZATION FOR THE DEPARTMENT OF CENSUS AND STATISTICS OF SRI LANKA**

1. IT 22266750 : NLP-Driven Identifier & Quasi-Identifier Detection
2. IT22364760 : Automatic SDC Pipeline Selection in Census Microdata While Balancing privacy-Utility trade offs (no model training or implementation required)
3. IT22129680: Integrating Differential Privacy with The Permutation Paradigm for Synthetic Data Generation
4. IT22319074: Evaluating Re-identification Risks in Anonymized Data Using Machine Learning

		IT22364760	IT 22266750	IT22129680	IT 22319074
Data Pipeline	**Data Sources:**	Full dataset handed by department of census and statistics, sent by email.	• Full dataset handed by department of census and statistics, sent by email. • publicly available datasets used for identification experimentation • Dataset contains quasi-identifiers such as age, gender, location, education, occupation, and other demographic attributes.	• Real microdata provided by the Department of Census and Statistics for research-based synthetic data generation. • Publicly available datasets used for privacy-safe experimentation, testing, and demonstration. • Dataset contains quasi-identifiers such as age, gender, location, education, occupation, and other demographic attributes. • Direct identifiers such as name, national ID, contact number, or exact address are removed before processing.	• Full dataset handed by department of census and statistics, sent by email. • publicly available datasets used for privacy-safe experimentation • Dataset contains quasi-identifiers such as age, gender, and location

			• The data is structured and contains multiple attributes	• Sensitive attributes are identified separately to ensure that confidential information is protected during synthetic data generation. • Data sources are selected to support structured microdata analysis, privacy preservation, and statistical utility evaluation.	• Direct identifiers (name, ID) removed prior to analysis
	Data Preprocessing:	• Remove structural identifiers • Considering empty cells as -1 and for text versions Treat as a category (“Unknown”) • Domain generalization hierarchy for Age and location	• Standardized column names, handled missing values, and corrected data types to ensure consistency • dentified and handled missing values using appropriate placeholder values • Verified and corrected data types (numeric and categorical) • Performed column profiling using unique value counts and percentage of uniqueness	• Structural identifiers and unnecessary indexing columns are removed before analysis. • Columns are classified into direct identifiers, quasi-identifiers, sensitive attributes, and non-sensitive attributes. • Missing numerical values are handled using placeholder values such as -1 where appropriate. • Missing categorical or text values are treated as a separate category such as **“Unknown”** to preserve data structure. • Duplicate records and inconsistent entries are cleaned to improve dataset quality. • Domain generalization hierarchies are applied for selected attributes such as age and location. • Age values may be grouped into ranges to reduce identifiability.	• Data cleaning performed to remove duplicates and inconsistent records • Missing values handled using imputation techniques • Categorical features encoded using label encoding / one-hot encoding • Numerical features normalized for improved model performance • Dataset structured into features (quasi-identifiers) and target (sensitive attribute)

		• Identified candidate quasi-identifiers based on data distribution • Prepared data for risk analysis by enabling evaluation of multi-column combinations (age + district + occupation)	• Location values may be generalized from specific areas to broader regions or districts. • Categorical attributes are encoded where required for processing and evaluation. • Numerical attributes are checked for valid ranges and normalized only where needed. • The dataset is prepared for synthetic generation while preserving important statistical patterns.	
Data storage and versioning	Datasets and Anonymized versions are stored in individual PCs and stored on protected google drive and one drive. For Data Versioning • Git (for code + configs) • File version naming (for datasets) • Metadata files (for reproducibility)	• Input datasets and processed outputs (risk reports, profiling results, QI combinations) are handled in CSV format • Basic version control is maintained using file naming conventions for datasets and outputs • Intermediate results (used for validation and analysis • Supports reproducibility and availability of results for evaluation	• Original datasets, processed datasets, and synthetic datasets are stored separately to avoid confusion. • Real datasets are stored securely on protected local machines, Google Drive, or OneDrive with restricted access. • Synthetic and anonymized outputs are stored in CSV format for easy testing, comparison, and reproducibility. • Dataset versions are maintained using clear file naming conventions such as `raw_data_v1`, `processed_data_v1`, and `synthetic_data_v1`. • Git is used for managing code, configuration files, and experiment scripts. • Metadata files are maintained to record dataset details, preprocessing steps,	• Processed datasets stored in CSV format for easy access and reproducibility • Version control maintained for datasets and preprocessing scripts • Data organized into training and testing subsets • Optional use of local storage or cloud storage for dataset management

				privacy settings, and generation parameters. • Experiment logs are maintained to track changes in privacy budget, noise level, permutation strategy, and evaluation results. • Version control supports reproducibility and makes it easier to compare different synthetic data generation outputs.	
Model Development	**Model Selection**	NOT APPLICABLE	• Selected a hybrid approach combining rule-based detection and data-driven analysis • Used Regex-based rules to detect structured identifiers (phone number patterns, ID-like formats) • Relied on column-level statistical features (uniqueness %, data type, column name) to identify quasi-identifiers and sensitive attributes • Chose this approach because census microdata is structured, and rule-based + statistical methods are more practical than complex models	• Traditional predictive model selection is not directly applicable because the main focus is synthetic data generation rather than classification or prediction.	• Random Forest Classifier selected for its robustness and ability to handle complex feature interactions • Gradient Boosting Classifier selected for high predictive accuracy and efficiency • Models chosen based on performance on structured tabular datasets • Focus on supervised learning approach for re-identification simulation

			• Focused on integrating detection with risk analysis measures (e.g., k-anonymity, uniqueness)		
	Model Training	NOT APPLICABLE	• Implemented detection logic using Regex-based rules for identifying structured identifiers • Applied column-level analysis using features such as uniqueness percentage and data distribution • Processed dataset columns to classify them into quasi-identifiers and sensitive attributes based on defined criteria • Validated detection results using risk analysis measures (e.g., k-anonymity, uniqueness)	• Conventional model training is not applicable, as the framework does not train a predictive model	• Dataset split into training and testing sets (e.g., 80:20 ratio) • Models trained using quasi-identifiers as input features • Target variable defined as sensitive attribute or identity proxy • Model performance evaluated on unseen test data • No deep learning models used to maintain simplicity and interpretability
Model Integration	**Tools used**	NOT APPLICABLE	• Backend development is carried out using Python • Pandas is used for data loading, preprocessing, and manipulation • NumPy is used for basic numerical operations	• Backend development is carried out using Python. • Pandas is used for data loading, cleaning, preprocessing, and manipulation. • NumPy is used for numerical operations and noise generation. • Differential privacy mechanisms are implemented using Python-based	• Backend: Python • Libraries: Scikit-learn, Pandas, NumPy • Data processing: Pandas for data manipulation • Model building: Scikit-learn

			• Regex (Regular Expressions) are used for rule-based identifier detection • Streamlit is used to build a simple interactive interface for uploading datasets and viewing analysis results • Jupyter Notebook and VS Code are used for development, testing, and experimentation	functions and privacy-related libraries where required. • Streamlit is used to build an interactive prototype interface for uploading datasets and generating synthetic outputs. • Matplotlib and Seaborn are used for visualizing data distributions, comparison charts, and evaluation results. • JSON codebooks are used to support value labels, categorical mappings, and metadata handling. • Jupyter Notebook and VS Code are used for development, testing, and experimentation.	• Development environment: Jupyter Notebook / VS Code • Visualization: Matplotlib / Seaborn (for evaluation graphs)
Model Deployment	**Testing Environments**	R Studio and Python Jypter notebook	• Tested in a local development environment using Python, with Jupyter Notebook and VS Code for experiments • Validated using real census data and public data • Evaluated different attribute combinations to verify detection and risk analysis behaviour • Assessed outputs based on correct identification of	• The system is tested in a local development environment using Python and Jupyter Notebook. • Different datasets are used to test whether the framework works on various microdata structures. • Testing is performed on both public datasets and privacy-safe sample datasets. • The synthetic output is validated by comparing it with the original dataset using utility and privacy metrics. • Evaluation includes distribution comparison, missingness comparison,	• Model tested on local development environment • Performance validated using test dataset • Evaluation metrics used to verify model reliability

			quasi-identifiers and sensitive attributes • Performed risk validation using k-anonymity and uniqueness %, including high-risk cases (k = 1) to confirm system effectiveness	correlation analysis, and nearest-neighbor privacy proxy analysis. • The framework is tested with different privacy budget values to observe the trade-off between privacy and utility.	
	Deployment Platform		• The system runs on a local machine environment for testing and demonstration • A Streamlit-based interface is used to upload datasets and display detection and risk analysis results • Users can analyze datasets and view outputs such as identified attributes and risk measures • Model outputs are generated as risk analysis results (e.g., k-anonymity, uniqueness %) and reports	• The system runs on a local machine environment for research demonstration and testing. • Users can upload a dataset, configure privacy settings, generate synthetic data, and view evaluation results. • Generated synthetic datasets and reports can be downloaded from the application. • Optional future deployment may include hosting the prototype on a secure internal server or private cloud environment. • The deployment is designed mainly for academic research, controlled experimentation, and privacy-safe data sharing demonstrations.	• System executed on local machine environment • Optional deployment using Flask or simple web interface • Model outputs generated as risk scores

	Deployment Method		• Input datasets are uploaded in CSV format through the Streamlit interface • The system performs data preprocessing, column profiling, and identifier detection using rule-based and statistical methods • Detected attributes are analyzed to identify quasi-identifiers and sensitive attributes • Risk analysis is performed using k-anonymity and uniqueness percentage on selected attribute combinations • Outputs are generated as risk analysis results and reports, which can be viewed through the interface	• .Input datasets are uploaded in CSV format. • The system performs column identification, preprocessing, privacy transformation, and synthetic data generation. • Differential privacy noise and permutation-based methods are applied during the generation process. • Synthetic datasets are generated and exported as CSV files. • Evaluation reports are produced to compare real and synthetic data. • Output includes synthetic data, utility metrics, privacy indicators, and visual comparison graphs. • The system does not require a trained machine learning model to be saved using pickle or joblib, as the process is generation-based rather than prediction-based.	• Trained model saved using joblib or pickle • Model loaded during runtime for prediction • Input dataset processed and passed to trained model • Output generated as re-identification risk level (High / Medium / Low)

Future Enhancements	**Model Improvement:**		• Improve detection accuracy using larger and more diverse datasets, including multilingual support (Sinhala & Tamil) • Enhance identification of hidden quasi-identifiers using improved data patterns and NLP-assisted methods • Develop a more interactive Streamlit dashboard with real-time visualization of risk metrics (k-anonymity, uniqueness %) • Enable user feedback and improved report generation with downloadable outputs • Improve analysis of multi-column relationships and external data linkage scenarios for better risk evaluation • Extend the system into a secure web-based platform with integration to tools such as ARX or sdcMicro	• Improve adaptive privacy budget allocation based on attribute sensitivity and analytical importance. • Enhance the permutation mechanism to better preserve relationships between variables. • Introduce advanced generative models such as GANs, VAEs, or diffusion models for more complex synthetic data generation. • Improve support for mixed datasets containing numerical, categorical, ordinal, and text-based attributes. • Strengthen privacy evaluation using advanced disclosure risk metrics such as membership inference risk and attribute inference risk. • Add automated detection of quasi-identifiers and sensitive attributes using rule-based and AI-assisted methods. • Improve utility preservation by maintaining correlations, subgroup patterns, and multivariate relationships. • Add a more detailed reporting module with downloadable privacy and utility evaluation reports. • Extend the system into a secure web-based platform for authorized users.	

				• Include role-based access control and audit logs for safer handling of sensitive datasets. • Validate the framework using larger real-world census-style datasets. • Compare the proposed APEDP approach with existing anonymization and synthetic data generation methods.	